

BAILING AGROCHEMICAL CO.,LTD.

江 苏 百 灵 农 化 有 限 公 司

**ADD: THE 4TH BRIDGE JIANGGUAN ROAD,JIANGYAN
JIANGSU CHINA**

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MATERIAL SAFETY DATA SHEET

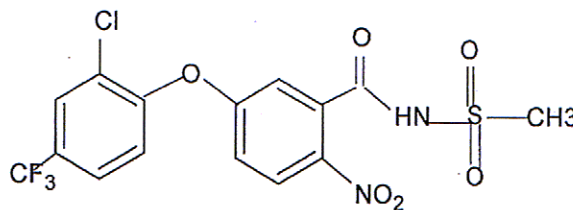
FOMESAFEN 25% SL

1 Product and Company Identification

Product Name: Fomesafen

Chemical Name: 5-[2-chloro-4-(trifluoromethyl)phenoxy]-N-(methylsulfonyl)-2-nitrobenzamide, sodium salt

Molecular structure:



Molecular Formula: C₁₅H₁₀ClF₃N₂O₆S

Molecular Weight: 438.76

Form: Liquid

Color: Brown

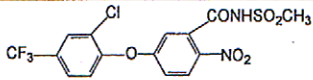
CAS No.: 108731-70-0

Manufacturer: BAILING AGROCHEMICAL CO.,LTD.

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2. Composition / Information On Ingredients

Composition	CAS No.	Content %	Molecular formula	Molecular weight
Fomesafen	72178-02-0	25.0min		438.76
30% liquid caustic	-----	7.0%max	-----	-----
Surfactant CL-18B	-----	5.0%max	-----	-----
Water	-----	Up to 100%	-----	-----

3. Hazards Identification

Classification of risk: WHO III; EPA II.

Potential health effects: Causes eye and skin irritation. It is toxicity if swallowed or absorbed through skin. Irritate to eyes and skin. May causes skin sensitization if contact skin.

Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

Unusual Fire, Explosion and Reactivity Hazards: During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

4. First Aid Measures

Have the product container, label or Material Safety Data Sheet with you when calling Qingdao Hansen, or a poison control center or doctor, or going for treatment.

If swallowed:

Call a poison control center or doctor immediately for treatment advice.

Have the person sip a glass of water if able to swallow.

Do not induce vomiting unless told to do so after calling a poison control center or doctor.

Do not give anything by mouth to an unconscious person.

If in eyes:

Hold eye open and rinse slowly and gently with water for 15-20 minutes.

Remove contact lenses, if present, after 5 minutes, then continue rinsing.

Call a poison control center or doctor for treatment advice.

If on skin or clothing:

Take off contaminated clothing.

Rinse skin immediately with plenty of water for 15-20 minutes.

Call a poison control center or doctor for treatment advice.

If inhaled:

Move person to fresh air.

If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible.

Call a poison control center or doctor for further treatment advice.

Notes to Physician: There is no specific antidote if this product is ingested. Treat symptomatically. Probable mucosal damage may contraindicate gastric lavage.

5. Fire-Fighting Measures

Fire and Explosion: Flash Point (Test Method): > 212°F

Unusual Fire, Explosion and Reactivity Hazards: During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire: Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

6. Accidental Release Measures

In Case of Spill or Leak: Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of

water. Clean up spills immediately, observing precautions outlined in Section 8. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent. Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

7. Handling and Storage

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

8. Exposure Controls/Personal Protection

Ingestion: prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

Eye Contact: Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Skin Contact: Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

Inhalation: Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. A NIOSH-certified combination air-purifying respirator with an N, P or R 95 or HE class filter and an organic vapor cartridge may be permissible under certain circumstances where airborne concentrations are expected to exceed exposure limits. Protection provided by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.

9. Physical and Chemical Properties

Appearance: Liquid (microemulsion)

Odor: Not determined

Boiling Point: 100.6°C

Flash point: not available

Specific Gravity/Density: 1.125 g/ml

pH: 6.0-9.0

10. Stability and Reactivity

Stability: Stable under normal use and storage conditions.

Hazardous Polymerization: Will not occur.

Conditions to Avoid: None known.

Materials to Avoid: Oxidizing agents.

Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

11. Toxicological Information

Acute Toxicity/Irritation Studies (Finished Product)

Ingestion: Slightly Toxic

Oral (LD50 Rat): > 3,863 mg/kg body weight

Dermal: Slightly Toxic

Dermal (LD50 Rat): > 2,000 mg/kg body weight

Inhalation: Practically Non-Toxic

Inhalation (LC50 Rat): > 2.1 mg/l air - 4 hours

Eye Contact: Moderately Irritating (Rabbit)

Skin Contact: Moderately Irritating (Rabbit)

Skin Sensitization: It is a moderate skin sensitizer in animal tests.

Reproductive/Developmental Effects-Fomesafen Sodium Salt:

Non-genotoxic in vitro and in vivo assays. Hepatic changes and a small effect on reproductive performance seen in a 3-generation rat study at the 1000 ppm dose level. (based on Fomesafen

Technical).

Chronic/Subchronic Toxicity Studies-Fomesafen Sodium Salt:

Liver effects seen at 10 ppm and 100 ppm in 90-day rat feeding studies and in 6-month dog studies at 25 mg/kg/d. (based on Fomesafen Technical). No evidence of neurotoxicity from subacute or longer-term studies in mammals. (based on Fomesafen Technical)

Carcinogenicity-Fomesafen Sodium Salt:

Increased rates of malignant liver tumors in a 2-yr mouse feeding study (1000 ppm), but the results are not considered relevant to man. (based on Fomesafen Technical).

Toxicity of Other Components

May cause severe eye irritation, possible burns. Causes skin irritation. Ingestion causes gastrointestinal irritation. May cause central nervous system depression, characterized by excitement, followed by headache, dizziness, drowsiness and nausea. Inhalation of high concentrations may cause central nervous system effects.

Target Organs

Active Ingredients

Fomesafen Sodium Salt: Liver (based on Fomesafen Technical)

Inert Ingredients

Cyclohexanol: Eye, skin, gastrointestinal tract, CNS

12. Ecological And Ecotoxicological Information

Summary of Effects

Fomesafen Sodium Salt: Toxic to fish. (based on Fomesafen Technical)

Eco-Acute Toxicity

Fomesafen Sodium Salt:

Rainbow Trout 96-hour LC_{50} 170 mg/L

Bluegill Sunfish 96-hour LC_{50} 1,507 mg/L

Mallard Oral LD_{50} >5,000 mg/kg

Mallard 5-day Dietary LC_{50} >20,000 mg/kg

Daphnia magna 48-hour EC_{50} 0.33 g/L

Green Algae EC_{50} 170 ug/L

Bobwhite Quail 5-day LC₅₀ >20,000 mg/kg

The above Ecotoxicity Data are based on Fomesafen Technical, not Fomesafen

Sodium Salt

Eco-Chronic Toxicity

Fomesafen SodiumSalt: Not Available

Environmental Fate

Fomesafen Sodium Salt:

The information presented here is for the active ingredient, fomesafen.

Persistent in soil. Stable in water. Highly mobile in soil. Will leach. Sinks in water (after 24 h).

13. Disposal Considerations

Disposal: Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.

Characteristic Waste: Not Applicable

Listed Waste: Not Applicable

14. Transport Information

Not Classified as Dangerous for Transport.

15. Regulatory Information

Not applicable

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipients

subsequently produce formulations containing this product, it is the recipient sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.

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